

Perfect  continuing same **exact format (PDF style + find → replace only, aligned to your code)**

CURRENT STATUS

✓ **Completed:** Day 1 → Day 23 

WHERE YOU ARE

System is powerful 

But NOT reliable 

👉 Problems:

- Excel query fails → system breaks
 - RAG weak → still uses bad result
 - No retry
 - No correction
-

WHAT IS NEXT Day 24 → SELF-CORRECTING AGENT

DAY 24 — SELF-CORRECTING AGENT (RETRY + FIX)

GOAL

Upgrade from: Fail once → stop 

to: Fail → retry → improve 

WHAT YOU ADD

Retry mechanism (max 2–3 attempts) + error feedback to LLM

STEP 1 — FIX EXCEL AGENT (RETRY LOOP): tools/excel_agent.py

FIND THIS BLOCK

```
result = eval(code, {"builtins": {}}, {"df": self.df, "pd": pd, "np": np})  
  
return str(result)
```

✅ REPLACE WITH

```
last_error = None
```

```
for attempt in range(3):
```

```
    error_hint = f"\nPrevious error: {last_error}" if last_error else ""
```

```
    prompt = f"""
```

You are a pandas code generator.

DataFrame: df

Schema:

{schema_desc}

Rules:

- Return ONLY python code

- No explanation

Question:

{question}

{error_hint}

```
"""
```

```
code = llm_func(prompt)
```

```
try:
```

```
    result = eval(code, {"builtins": {}}, {"df": self.df, "pd": pd, "np": np})
```

```
    return str(result)
```

```
except Exception as e:
```

```
    last_error = str(e)
```

```
return f"Failed after retries: {last_error}"
```

✅ STEP 2 — FIX RAG AGENT (LOW QUALITY RETRY): app/rag_agent.py

 **FIND:** return generate(prompt)

✅ REPLACE WITH

```
answer = generate(prompt)

for _ in range(2):

    if "not in context" in answer.lower() or len(answer.strip()) < 20:

        fix_prompt = f"""

The previous answer was weak.

Context:

{context}

Question:

{query}

Improve the answer.

"""

        answer = generate(fix_prompt)

return answer
```

✅ ✅ STEP 3 — FINAL ANSWER RETRY (MAIN AGENT): app/agent.py

 **FIND:** final_answer = generate(final_prompt)

✅ REPLACE WITH

```
final_answer = generate(final_prompt)

# ✅ retry if weak

for _ in range(2):

    if len(final_answer.strip()) < 30:

        fix_prompt = f"""Improve this answer:

{final_answer}

Question:

{query}"""

        final_answer = generate(fix_prompt)
```

✅✅ STEP 4 — OPTIONAL LOG (GOOD PRACTICE): 📄 app/agent.py

✅ ADD AFTER RETRY

```
if len(final_answer.strip()) < 30:  
    print("[WARNING] Answer still weak after retry")
```

✅✅ FINAL FLOW NOW

User Query

↓

Planner

↓

Agents (Excel / RAG)

↓

⚡ Retry if failure ✅

↓

Final Answer

↓

⚡ Retry if weak ✅

✅✅ TEST

Run: `uvicorn app.api:app --reload`

✅ Try:

>> invalid excel query

>> ambiguous document question

>> random vague input

✅ EXPECTED

Before: fails ❌

Now: retries + improves 

  **WHAT YOU BUILT:** Self-correcting AI agent

  **SYSTEM LEVEL NOW**

 AI system

 Multi-agent

 Memory

 Evaluation

 Monitoring

  Self-correction (NEW)

  **WHY THIS IS BIG**

Now your system:

- handles failures
 - improves outputs
 - becomes robust
 - behaves like real Copilot
-

  **INDUSTRY TERM**

Self-healing / Autonomous agent loop

Used in:

- AutoGPT
 - LangGraph advanced
 - Copilot systems
-

  **NEXT (FINAL)**

Day 25 → DEPLOYMENT (cloud / production setup)

If ready:  say **Day 25** 